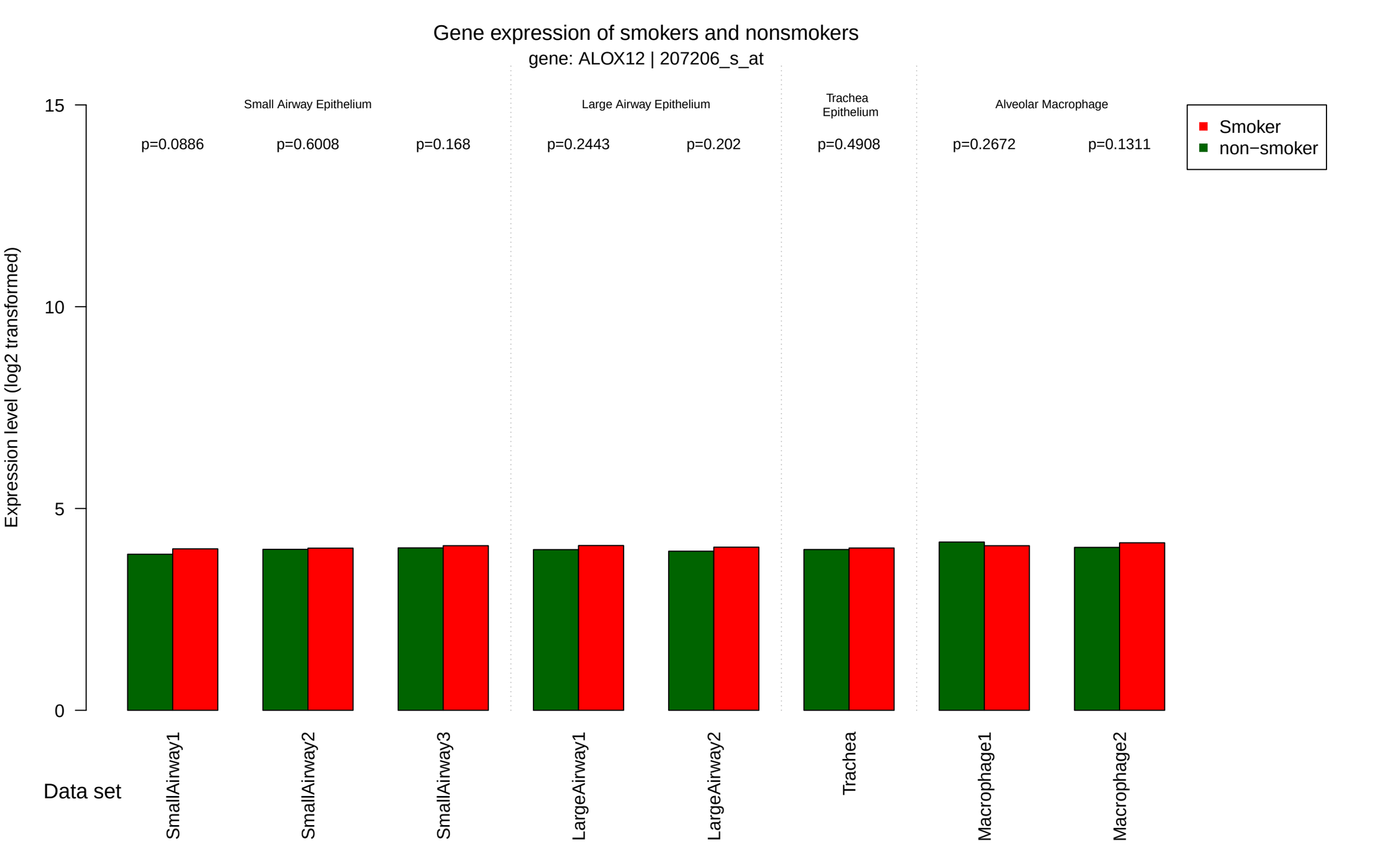
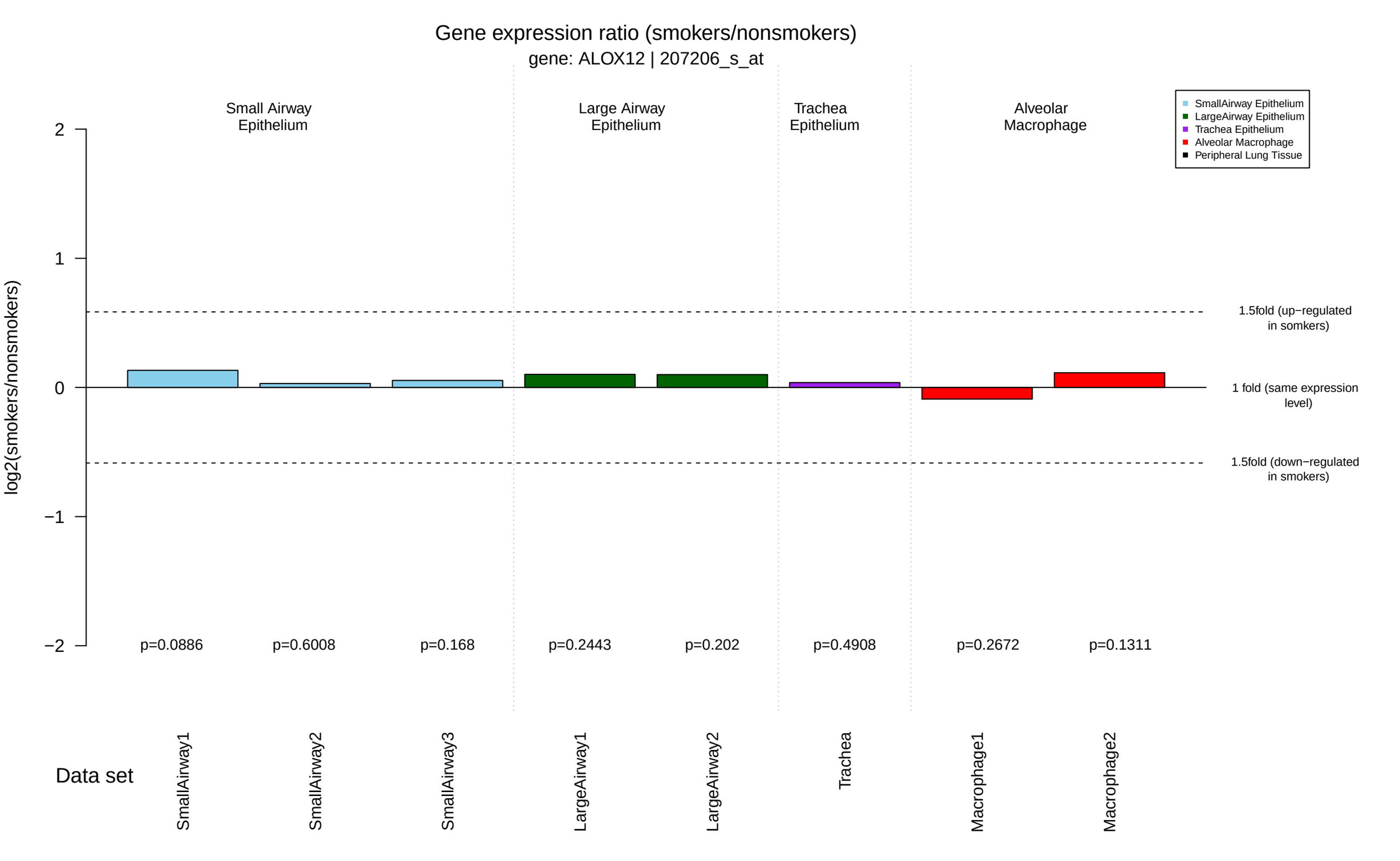
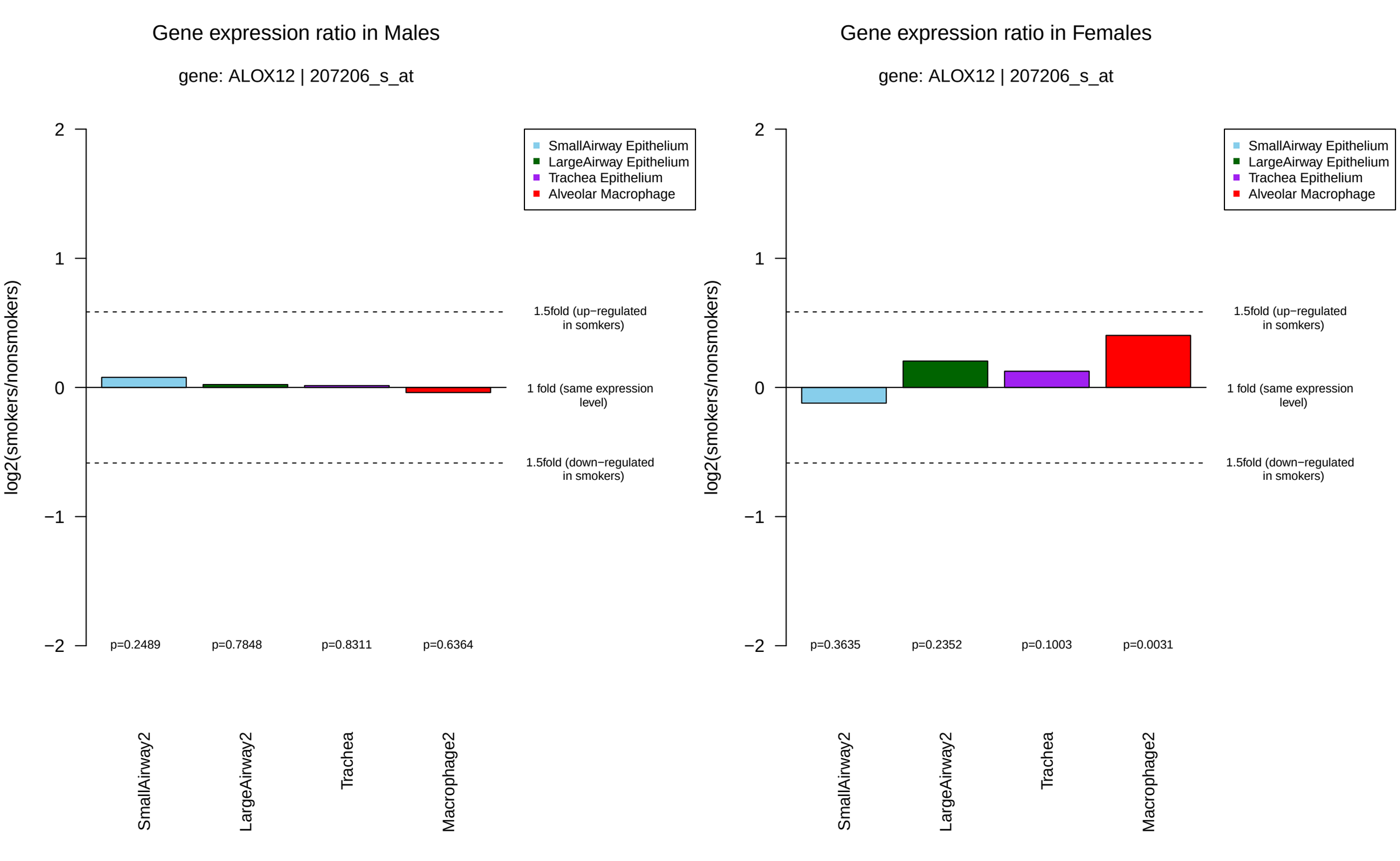
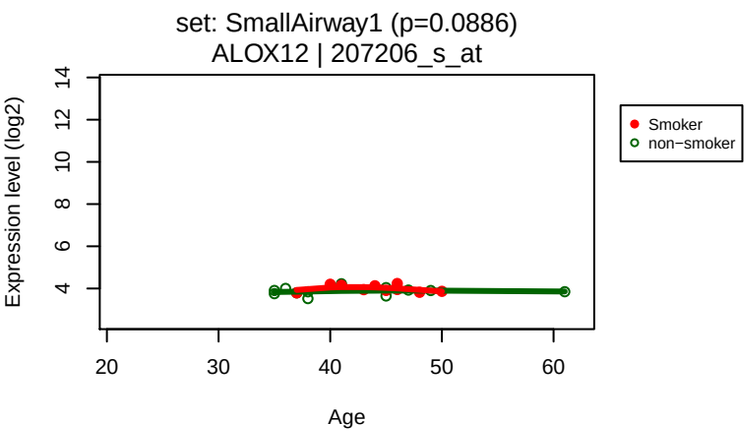
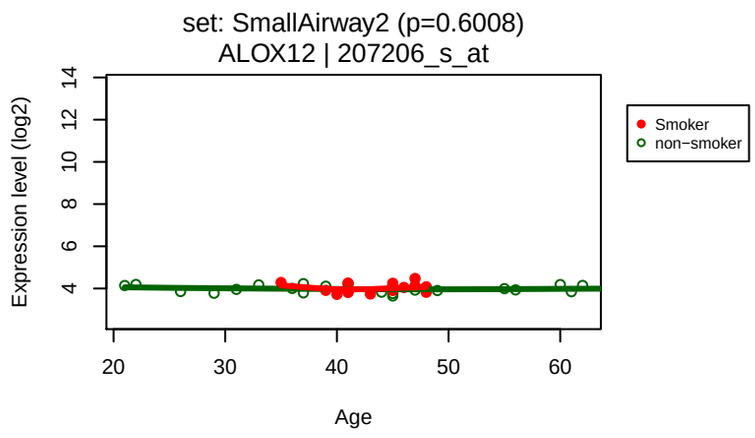
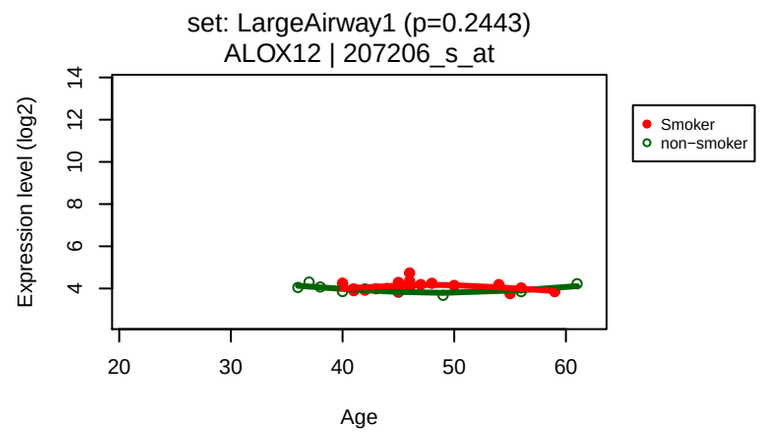

Gene expression of smokers and nonsmokers
gene: ALOX12 | 207206_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0886
p=0.6008
p=0.168
p=0.2443
p=0.202
p=0.4908
p=0.2672
p=0.1311
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

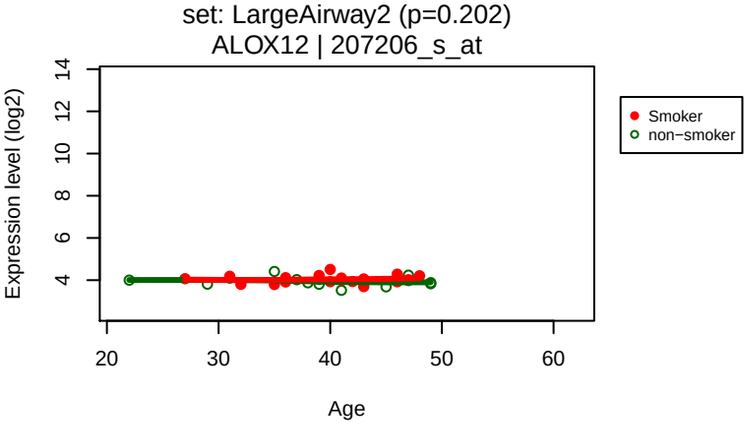

Gene expression ratio (smokers/nonsmokers)
gene: ALOX12 | 207206_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0886
p=0.6008
p=0.168
p=0.2443
p=0.202
p=0.4908
p=0.2672
p=0.1311
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

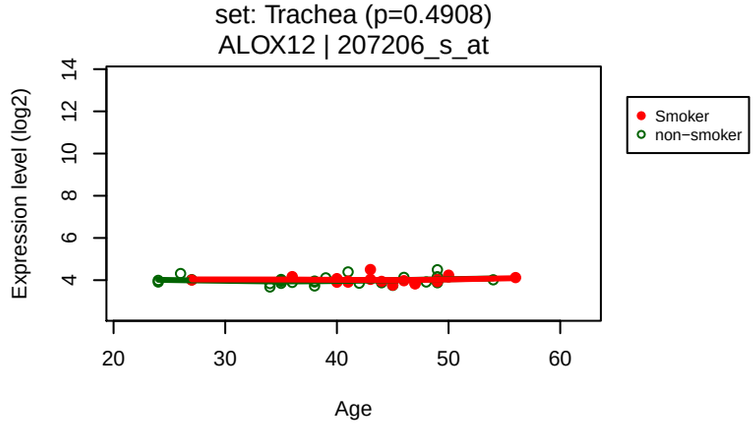

Gene expression ratio in Males
gene: ALOX12 | 207206_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2489
p=0.7848
p=0.8311
p=0.6364
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALOX12 | 207206_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3635
p=0.2352
p=0.1003
p=0.0031
SmallAirway2
LargeAirway2
Trachea
Macrophage2

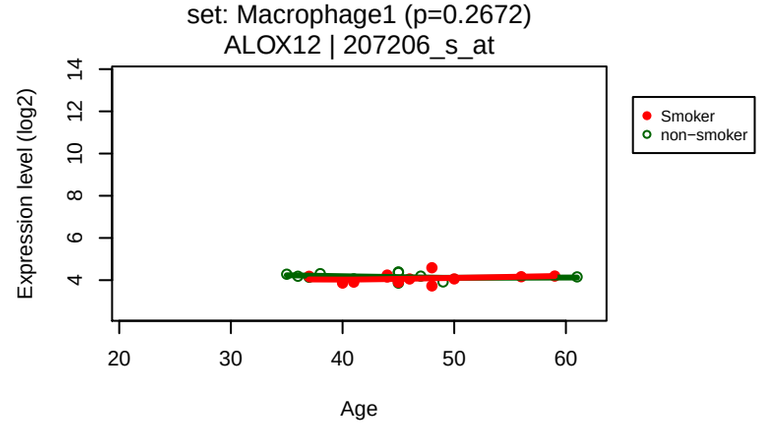

set: SmallAirway1 (p=0.0886)
ALOX12 | 207206_s_at
Expression level (log2)
Age
Smoker
non−smoker

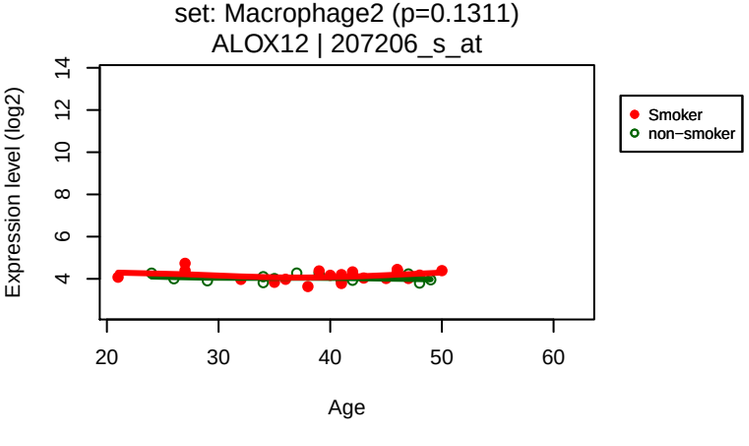

set: SmallAirway2 (p=0.6008)
ALOX12 | 207206_s_at
Expression level (log2)
Age
Smoker
non−smoker

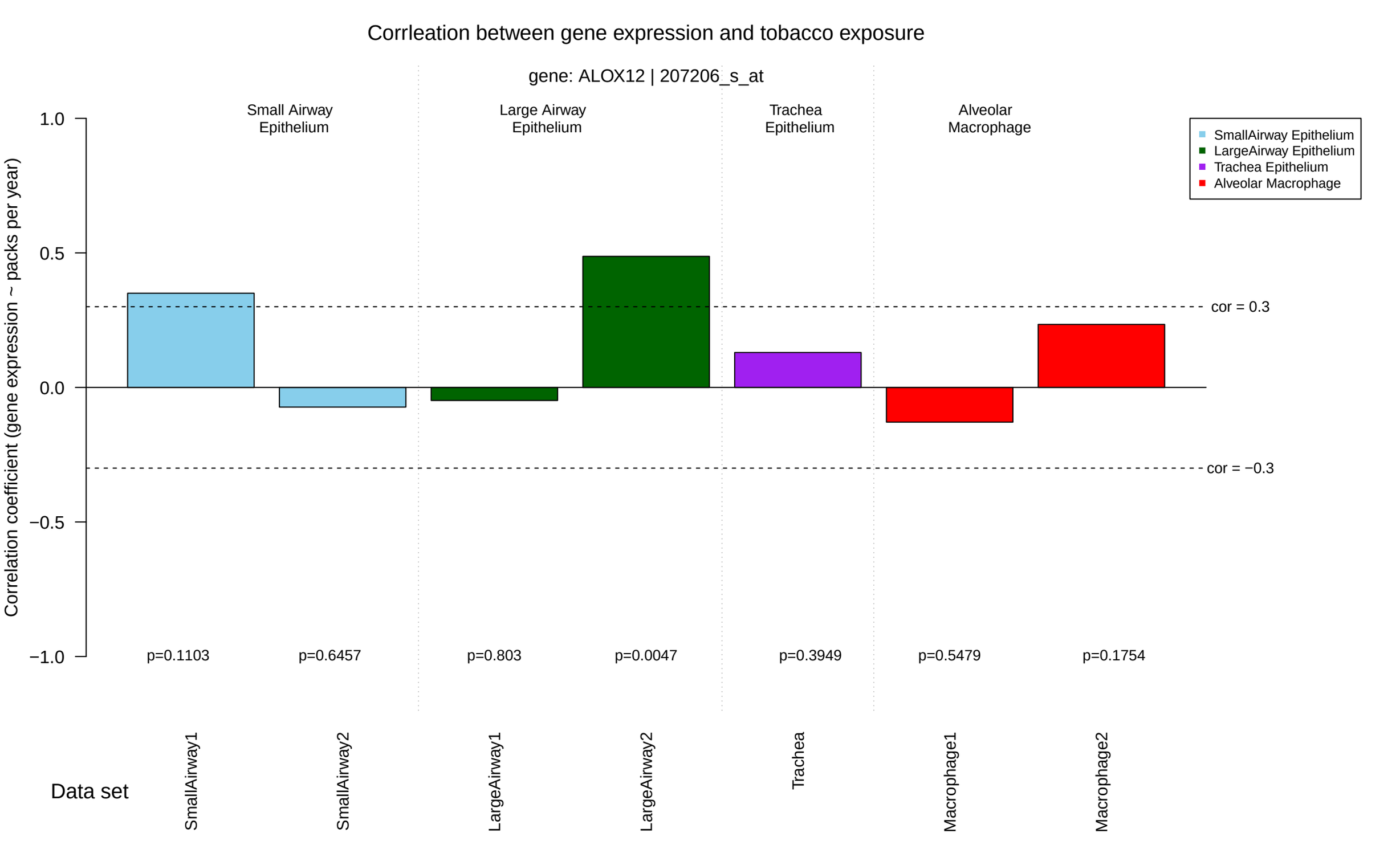

set: LargeAirway1 (p=0.2443)
ALOX12 | 207206_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.202)
ALOX12 | 207206_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4908)
ALOX12 | 207206_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2672)
ALOX12 | 207206_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1311)
ALOX12 | 207206_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALOX12 | 207206_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1103
p=0.6457
p=0.803
p=0.0047
p=0.3949
p=0.5479
p=0.1754
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2